

Module 22 – Naive Bayes Case Study on Titanic Dataset

Objective

In this module, you will apply the **Naive Bayes** classification algorithm to build a **Titanic Survival Prediction Model**. This case study will help you understand probabilistic classification and how to implement Naive Bayes for real-world datasets.

Reference

Watch the **Module 22 video** carefully. Follow the concepts and implementation steps explained in the video to complete your case study.

Dataset

You can use the **Titanic dataset** available on [Kaggle](#).
Search for “**Titanic - Machine Learning from Disaster**” dataset.

Task Instructions

1. Import and explore the dataset (inspect columns, missing values, and data types).
2. Perform data cleaning and preprocessing:
 - Handle missing values
 - Encode categorical variables (e.g., Sex, Embarked)
3. Split the dataset into **training** and **testing** sets.
4. Implement the **Naive Bayes classifier** to predict passenger survival.
5. Evaluate the model using metrics like **accuracy**, **confusion matrix**, and **classification report**.
6. Provide a brief analysis of your findings and insights.

Submission Requirements

- Submit your **.ipynb** file (Jupyter Notebook).
- Upload your notebook to **Google Colab** and share the **Colab link**.
- Ensure the link sharing is set to “**Anyone with the link can view**”.